CORE PURE / PROOF BY INDUCTION

Proof by Induction - Matrices

Worked solutions 8 questions · 55 marks

EXAM BOARDS Edexcel 9FM0 · AQA 7367 · OCR A H245 · OCR MEI H645 · CIE 9231

Questions

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1. Prove by induction that, for $n \in \mathbb{Z}^+$,

$$\begin{pmatrix} 2 & 1 \\ 0 & 3 \end{pmatrix}^n = \begin{pmatrix} 2^n & 3^n - 2^n \\ 0 & 3^n \end{pmatrix}$$

[6]

Solution

Let

$$\mathbf{A} = \begin{pmatrix} 2 & 1 \\ 0 & 3 \end{pmatrix}$$

We prove by induction that

$$\mathbf{A}^n = \begin{pmatrix} 2^n & 3^n - 2^n \\ 0 & 3^n \end{pmatrix} \quad \text{for all } n \in \mathbb{Z}^+$$

Base case: $n = 1$. The left-hand side is

$$\mathbf{A}^1 = \mathbf{A} = \begin{pmatrix} 2 & 1 \\ 0 & 3 \end{pmatrix}$$

The right-hand side when $n = 1$ is

$$\begin{pmatrix} 2^1 & 3^1 - 2^1 \\ 0 & 3^1 \end{pmatrix} = \begin{pmatrix} 2 & 1 \\ 0 & 3 \end{pmatrix}$$

So the statement is true for $n = 1$.

Inductive hypothesis: Assume the statement is true for $n = k$, where $k \in \mathbb{Z}^+$. That is, assume

$$\mathbf{A}^k = \begin{pmatrix} 2^k & 3^k - 2^k \\ 0 & 3^k \end{pmatrix}$$

Inductive step: We must show that this implies

$$\mathbf{A}^{k+1} = \begin{pmatrix} 2^{k+1} & 3^{k+1} - 2^{k+1} \\ 0 & 3^{k+1} \end{pmatrix}$$

Using $\mathbf{A}^{k+1} = \mathbf{A}^k \mathbf{A}$ and the inductive hypothesis:

$$\begin{aligned} \mathbf{A}^{k+1} &= \begin{pmatrix} 2^k & 3^k - 2^k \\ 0 & 3^k \end{pmatrix} \begin{pmatrix} 2 & 1 \\ 0 & 3 \end{pmatrix} \\ &= \begin{pmatrix} 2^k \times 2 + (3^k - 2^k) \times 0 & 2^k \times 1 + (3^k - 2^k) \times 3 \\ 0 \times 2 + 3^k \times 0 & 0 \times 1 + 3^k \times 3 \end{pmatrix} \\ &= \begin{pmatrix} 2^{k+1} & 2^k + 3(3^k - 2^k) \\ 0 & 3^{k+1} \end{pmatrix} \end{aligned}$$

Now simplify the top-right entry:

$$\begin{aligned} 2^k + 3(3^k - 2^k) &= 2^k + 3^{k+1} - 3 \times 2^k \\ &= 3^{k+1} - 2 \times 2^k \\ &= 3^{k+1} - 2^{k+1} \end{aligned}$$

So

$$\mathbf{A}^{k+1} = \begin{pmatrix} 2^{k+1} & 3^{k+1} - 2^{k+1} \\ 0 & 3^{k+1} \end{pmatrix}$$

This is exactly the required form, so the statement is true for $n = k + 1$.

Conclusion: If the statement is true for $n = k$, then it is true for $n = k + 1$. Since it is true for $n = 1$, by the principle of mathematical induction the statement is true for all $n \in \mathbb{Z}^+$.

2. Prove by mathematical induction that, for $n \in \mathbb{N}$,

$$\begin{pmatrix} 3 & -1 \\ 1 & 1 \end{pmatrix}^n = 2^{n-1} \begin{pmatrix} n+2 & -n \\ n & 2-n \end{pmatrix}$$

[6]

Solution

Let

$$\mathbf{A} = \begin{pmatrix} 3 & -1 \\ 1 & 1 \end{pmatrix}$$

We will prove by mathematical induction that

$$\mathbf{A}^n = 2^{n-1} \begin{pmatrix} n+2 & -n \\ n & 2-n \end{pmatrix} \qquad \text{for all } n \in \mathbb{N}$$

Base case: $n = 1$. The left-hand side is

$$\mathbf{A}^1 = \mathbf{A} = \begin{pmatrix} 3 & -1 \\ 1 & 1 \end{pmatrix}$$

Now evaluate the right-hand side when $n = 1$:

$$2^{1-1} \begin{pmatrix} 1+2 & -1 \\ 1 & 2-1 \end{pmatrix} = 1 \begin{pmatrix} 3 & -1 \\ 1 & 1 \end{pmatrix} = \begin{pmatrix} 3 & -1 \\ 1 & 1 \end{pmatrix}$$

So the statement is true for $n = 1$.

Inductive hypothesis: Assume the statement is true for $n = k$, where $k \in \mathbb{N}$. That is, assume

$$\mathbf{A}^k = 2^{k-1} \begin{pmatrix} k+2 & -k \\ k & 2-k \end{pmatrix}$$

Inductive step: We must show that this implies

$$\mathbf{A}^{k+1} = 2^k \begin{pmatrix} (k+1)+2 & -(k+1) \\ k+1 & 2-(k+1) \end{pmatrix}$$

Using $\mathbf{A}^{k+1} = \mathbf{A}^k \mathbf{A}$ and the inductive hypothesis:

$$\begin{aligned} \mathbf{A}^{k+1} &= \mathbf{A}^k \mathbf{A} \\ &= \left(2^{k-1} \begin{pmatrix} k+2 & -k \\ k & 2-k \end{pmatrix} \right) \begin{pmatrix} 3 & -1 \\ 1 & 1 \end{pmatrix} \\ &= 2^{k-1} \begin{pmatrix} k+2 & -k \\ k & 2-k \end{pmatrix} \begin{pmatrix} 3 & -1 \\ 1 & 1 \end{pmatrix} \end{aligned}$$

Now multiply the matrices:

$$\begin{aligned} \begin{pmatrix} k+2 & -k \\ k & 2-k \end{pmatrix} \begin{pmatrix} 3 & -1 \\ 1 & 1 \end{pmatrix} &= \begin{pmatrix} 3(k+2)+(-k)(1) & (k+2)(-1)+(-k)(1) \\ 3k+(2-k)(1) & k(-1)+(2-k)(1) \end{pmatrix} \\ &= \begin{pmatrix} 3k+6-k & -k-2-k \\ 3k+2-k & -k+2-k \end{pmatrix} \\ &= \begin{pmatrix} 2k+6 & -2k-2 \\ 2k+2 & 2-2k \end{pmatrix} \end{aligned}$$

So

$$\mathbf{A}^{k+1} = 2^{k-1}\begin{pmatrix} 2k+6 & -2k-2 \\ 2k+2 & 2-2k \end{pmatrix}$$

Factor out 2 from the matrix:

$$\mathbf{A}^{k+1} = 2^{k}\begin{pmatrix} k+3 & -(k+1) \\ k+1 & 1-k \end{pmatrix}$$

Since $1-k = 2-(k+1)$, this becomes

$$\mathbf{A}^{k+1} = 2^{k}\begin{pmatrix} k+3 & -(k+1) \\ k+1 & 2-(k+1) \end{pmatrix}$$

This is exactly the required form, so the statement is true for $n = k+1$.

Conclusion: If the statement is true for $n = k$, then it is true for $n = k+1$. Since it is true for $n = 1$, by the principle of mathematical induction the statement is true for all $n \in \mathbb{N}$.

3. The matrix $\mathbf{A}$ is given by

$$\mathbf{A} = \begin{pmatrix} 4 & 2 \\ -2 & 0 \end{pmatrix}$$

(a) Prove by induction that

$$\mathbf{A}^n = 2^n \begin{pmatrix} n+1 & n \\ -n & 1-n \end{pmatrix}$$

for all positive integers n. **[6]**

(b) The matrix $\mathbf{B}$ is given by

$$\mathbf{B} = (\mathbf{A}^n)^{-1}$$

Hence find $\mathbf{B}$ in terms of n. **[2]**

Solution

(a) We prove by induction that

$$\mathbf{A}^n = 2^n \begin{pmatrix} n+1 & n \\ -n & 1-n \end{pmatrix}$$

for all positive integers n.

Base case: $n = 1$. The claimed formula gives

$$\mathbf{A}^1 = 2^1 \begin{pmatrix} 2 & 1 \\ -1 & 0 \end{pmatrix} = \begin{pmatrix} 4 & 2 \\ -2 & 0 \end{pmatrix} = \mathbf{A}$$

So the statement is true for $n = 1$.

Inductive hypothesis: Assume the statement is true for $n = k$, where k is a positive integer. That is, assume

$$\mathbf{A}^k = 2^k \begin{pmatrix} k+1 & k \\ -k & 1-k \end{pmatrix}$$

Inductive step: We must show that

$$\mathbf{A}^{k+1} = 2^{k+1} \begin{pmatrix} k+2 & k+1 \\ -(k+1) & 1-(k+1) \end{pmatrix}$$

Using $\mathbf{A}^{k+1} = \mathbf{A}^k \mathbf{A}$ and the inductive hypothesis:

$$\begin{aligned} \mathbf{A}^{k+1} &= \mathbf{A}^k \mathbf{A} \\ &= 2^k \begin{pmatrix} k+1 & k \\ -k & 1-k \end{pmatrix} \begin{pmatrix} 4 & 2 \\ -2 & 0 \end{pmatrix} \end{aligned}$$

Now multiply the matrices:

$$\begin{aligned} \mathbf{A}^{k+1} &= 2^k \begin{pmatrix} 4(k+1) - 2k & 2(k+1) + 0 \\ -4k - 2(1-k) & -2k + 0 \end{pmatrix} \\ &= 2^k \begin{pmatrix} 2k+4 & 2k+2 \\ -2k-2 & -2k \end{pmatrix} \end{aligned}$$

Factor out 2 from the matrix:

$$\mathbf{A}^{k+1} = 2^{k+1} \begin{pmatrix} k+2 & k+1 \\ -(k+1) & -k \end{pmatrix}$$

Since

$$-k = 1 - (k+1)$$

this becomes

$$\mathbf{A}^{k+1} = 2^{k+1}\begin{pmatrix} k+2 & k+1 \\ -(k+1) & 1-(k+1) \end{pmatrix}$$

This is exactly the required form, so the statement is true for $n = k+1$.

Conclusion: If the statement is true for $n = k$, then it is true for $n = k+1$. Since it is true for $n = 1$, by the principle of mathematical induction the statement is true for all positive integers n.

(b) Let

$$\mathbf{M} = \begin{pmatrix} n+1 & n \\ -n & 1-n \end{pmatrix}$$

so that part (a) gives

$$\mathbf{A}^n = 2^n\mathbf{M}$$

Hence

$$\mathbf{B} = (\mathbf{A}^n)^{-1} = (2^n\mathbf{M})^{-1} = 2^{-n}\mathbf{M}^{-1}$$

Now find $\mathbf{M}^{-1}$. The inverse of a 2×2 matrix is

$$\begin{pmatrix} a & b \\ c & d \end{pmatrix}^{-1} = \frac{1}{ad-bc}\begin{pmatrix} d & -b \\ -c & a \end{pmatrix}$$

First calculate the determinant:

$$\begin{aligned} \det(\mathbf{M}) &= (n+1)(1-n) - n(-n) \\ &= (n+1)(1-n) + n^2 \\ &= 1 - n^2 + n^2 \\ &= 1 \end{aligned}$$

So

$$\mathbf{M}^{-1} = \begin{pmatrix} 1-n & -n \\ n & n+1 \end{pmatrix}$$

Therefore

$$\mathbf{B} = \frac{1}{2^n}\begin{pmatrix} 1-n & -n \\ n & n+1 \end{pmatrix}$$

4. The matrix $\mathbf{A}$ is given by

$$\mathbf{A} = \begin{pmatrix} 2 & 0 & 0 \\ 1 & 2 & 0 \\ 0 & 1 & 2 \end{pmatrix}$$

Prove by induction that, for $n \geq 1$,

$$\mathbf{A}^n = \begin{pmatrix} 2^n & 0 & 0 \\ n\,2^{n-1} & 2^n & 0 \\ \frac{n(n-1)}{2}\,2^{n-2} & n\,2^{n-1} & 2^n \end{pmatrix} \qquad \textbf{[6]}$$

Solution

We prove by induction that

$$\mathbf{A}^n = \begin{pmatrix} 2^n & 0 & 0 \\ n\,2^{n-1} & 2^n & 0 \\ \frac{n(n-1)}{2}\,2^{n-2} & n\,2^{n-1} & 2^n \end{pmatrix}$$

Base case: $n = 1$. The right-hand side is

$$\begin{pmatrix} 2^1 & 0 & 0 \\ 1 \times 2^0 & 2^1 & 0 \\ \frac{1(1-1)}{2}\,2^{-1} & 1 \times 2^0 & 2^1 \end{pmatrix} = \begin{pmatrix} 2 & 0 & 0 \\ 1 & 2 & 0 \\ 0 & 1 & 2 \end{pmatrix} = \mathbf{A}$$

So the statement is true for $n = 1$.

Inductive hypothesis: Assume the statement is true for $n = k$, where k is an integer with $k \geq 1$. That is, assume

$$\mathbf{A}^k = \begin{pmatrix} 2^k & 0 & 0 \\ k\,2^{k-1} & 2^k & 0 \\ \frac{k(k-1)}{2}\,2^{k-2} & k\,2^{k-1} & 2^k \end{pmatrix}$$

Inductive step: We must show that the statement is true for $n = k + 1$.

Using $\mathbf{A}^{k+1} = \mathbf{A}^k\mathbf{A}$ and the inductive hypothesis:

$$\begin{aligned} \mathbf{A}^{k+1} &= \begin{pmatrix} 2^k & 0 & 0 \\ k\,2^{k-1} & 2^k & 0 \\ \frac{k(k-1)}{2}\,2^{k-2} & k\,2^{k-1} & 2^k \end{pmatrix} \begin{pmatrix} 2 & 0 & 0 \\ 1 & 2 & 0 \\ 0 & 1 & 2 \end{pmatrix} \\ &= \begin{pmatrix} 2^k \times 2 & 0 & 0 \\ k\,2^{k-1} \times 2 + 2^k \times 1 & 2^k \times 2 & 0 \\ \frac{k(k-1)}{2}\,2^{k-2} \times 2 + k\,2^{k-1} \times 1 & k\,2^{k-1} \times 2 + 2^k \times 1 & 2^k \times 2 \end{pmatrix} \\ &= \begin{pmatrix} 2^{k+1} & 0 & 0 \\ k\,2^k + 2^k & 2^{k+1} & 0 \\ \frac{k(k-1)}{2}\,2^{k-1} + k\,2^{k-1} & k\,2^k + 2^k & 2^{k+1} \end{pmatrix} \\ &= \begin{pmatrix} 2^{k+1} & 0 & 0 \\ (k+1)2^k & 2^{k+1} & 0 \\ \left(\frac{k(k-1)}{2} + k\right)2^{k-1} & (k+1)2^k & 2^{k+1} \end{pmatrix} \end{aligned}$$

Now simplify the bottom-left entry:

$$\left(\frac{k(k-1)}{2}+k\right)2^{k-1} = \left(\frac{k(k-1)+2k}{2}\right)2^{k-1}$$
$$= \left(\frac{k^2-k+2k}{2}\right)2^{k-1}$$
$$= \frac{k(k+1)}{2}\,2^{k-1}$$

So

$$\mathbf{A}^{k+1} = \begin{pmatrix} 2^{k+1} & 0 & 0 \\ (k+1)2^k & 2^{k+1} & 0 \\ \frac{k(k+1)}{2}\,2^{k-1} & (k+1)2^k & 2^{k+1} \end{pmatrix}$$

This is exactly the required form with $n = k+1$, since

$$(k+1)2^k = (k+1)2^{(k+1)-1}$$

and

$$\frac{k(k+1)}{2}\,2^{k-1} = \frac{(k+1)\big((k+1)-1\big)}{2}\,2^{(k+1)-2}$$

so the statement is true for $n = k+1$.

Conclusion: If the statement is true for $n = k$, then it is true for $n = k+1$. Since it is true for $n = 1$, by the principle of mathematical induction the statement is true for all $n \geq 1$.

5. The matrix $\mathbf{M}$ is given by $\mathbf{M} = \begin{pmatrix} 3c & 1 \\ 0 & 3c \end{pmatrix}$, where c is a positive constant.

(a) Prove by induction that for all positive integers n,

$$\mathbf{M}^n = \begin{pmatrix} (3c)^n & n(3c)^{n-1} \\ 0 & (3c)^n \end{pmatrix}$$

[6]

(b) Given that $\det(\mathbf{M}^n) = 144^n$, find the value of c. **[5]**

Solution

(a) We prove the result by induction on n.

Base case: $n = 1$. The left-hand side is

$$\mathbf{M}^1 = \mathbf{M} = \begin{pmatrix} 3c & 1 \\ 0 & 3c \end{pmatrix}$$

The right-hand side when $n = 1$ is

$$\begin{pmatrix} (3c)^1 & 1(3c)^{1-1} \\ 0 & (3c)^1 \end{pmatrix} = \begin{pmatrix} (3c)^1 & 1(3c)^0 \\ 0 & (3c)^1 \end{pmatrix} = \begin{pmatrix} 3c & 1 \\ 0 & 3c \end{pmatrix}$$

So the statement is true for $n = 1$.

Inductive hypothesis: Assume the statement is true for $n = k$, where k is a positive integer. That is, assume

$$\mathbf{M}^k = \begin{pmatrix} (3c)^k & k(3c)^{k-1} \\ 0 & (3c)^k \end{pmatrix}$$

Inductive step: We must show that this implies

$$\mathbf{M}^{k+1} = \begin{pmatrix} (3c)^{k+1} & (k+1)(3c)^k \\ 0 & (3c)^{k+1} \end{pmatrix}$$

Using $\mathbf{M}^{k+1} = \mathbf{M}^k\mathbf{M}$ and the inductive hypothesis:

$$\begin{aligned} \mathbf{M}^{k+1} &= \mathbf{M}^k\mathbf{M} \\ &= \begin{pmatrix} (3c)^k & k(3c)^{k-1} \\ 0 & (3c)^k \end{pmatrix}\begin{pmatrix} 3c & 1 \\ 0 & 3c \end{pmatrix} \\ &= \begin{pmatrix} (3c)^k \times 3c + k(3c)^{k-1} \times 0 & (3c)^k \times 1 + k(3c)^{k-1} \times 3c \\ 0 \times 3c + (3c)^k \times 0 & 0 \times 1 + (3c)^k \times 3c \end{pmatrix} \\ &= \begin{pmatrix} (3c)^{k+1} & (3c)^k + 3ck(3c)^{k-1} \\ 0 & (3c)^{k+1} \end{pmatrix} \end{aligned}$$

Now simplify the top-right entry:

$$\begin{aligned} (3c)^k + 3ck(3c)^{k-1} &= (3c)^k + k(3c)^k \\ &= (k+1)(3c)^k \end{aligned}$$

Hence

$$\mathbf{M}^{k+1} = \begin{pmatrix} (3c)^{k+1} & (k+1)(3c)^k \\ 0 & (3c)^{k+1} \end{pmatrix}$$

This is exactly the required form, so the statement is true for $n = k + 1$.

Conclusion: If the statement is true for $n = k$, then it is true for $n = k + 1$. Since it is true for $n = 1$, by the principle of mathematical induction the statement is true for all positive integers n.

(b) Part (a) gives

$$\mathbf{M}^n = \begin{pmatrix} (3c)^n & n(3c)^{n-1} \\ 0 & (3c)^n \end{pmatrix}$$

Its determinant is the product of the leading diagonal entries minus the product of the other two, so

$$\begin{aligned} \det(\mathbf{M}^n) &= (3c)^n \times (3c)^n - n(3c)^{n-1} \times 0 \\ &= (3c)^{2n} \\ &= (9c^2)^n \end{aligned}$$

We are given that

$$\det(\mathbf{M}^n) = 144^n$$

So

$$(9c^2)^n = 144^n$$

Hence

$$9c^2 = 144$$

Therefore

$$c^2 = 16$$

Since c is positive, taking the positive root gives

$$c = 4$$

6. For $n \in \mathbb{Z}^+$, show, using mathematical induction, that

$$\begin{pmatrix} -1 & 1 & 0 \\ 0 & -1 & 1 \\ 0 & 0 & -1 \end{pmatrix}^n = \begin{pmatrix} (-1)^n & -n(-1)^n & \frac{n(n-1)}{2}(-1)^n \\ 0 & (-1)^n & -n(-1)^n \\ 0 & 0 & (-1)^n \end{pmatrix} \quad \textbf{[6]}$$

Solution

Let

$$\mathbf{A} = \begin{pmatrix} -1 & 1 & 0 \\ 0 & -1 & 1 \\ 0 & 0 & -1 \end{pmatrix}$$

We prove by induction that

$$\mathbf{A}^n = \begin{pmatrix} (-1)^n & -n(-1)^n & \frac{n(n-1)}{2}(-1)^n \\ 0 & (-1)^n & -n(-1)^n \\ 0 & 0 & (-1)^n \end{pmatrix}$$

Base case: $n = 1$. Substituting $n = 1$ into the right-hand side gives

$$\begin{pmatrix} (-1)^1 & -1(-1)^1 & \frac{1(1-1)}{2}(-1)^1 \\ 0 & (-1)^1 & -1(-1)^1 \\ 0 & 0 & (-1)^1 \end{pmatrix} = \begin{pmatrix} -1 & 1 & 0 \\ 0 & -1 & 1 \\ 0 & 0 & -1 \end{pmatrix} = \mathbf{A}$$

So the statement is true for $n = 1$.

Inductive hypothesis: Assume the statement is true for $n = k$, where $k \in \mathbb{Z}^+$. That is, assume

$$\mathbf{A}^k = \begin{pmatrix} (-1)^k & -k(-1)^k & \frac{k(k-1)}{2}(-1)^k \\ 0 & (-1)^k & -k(-1)^k \\ 0 & 0 & (-1)^k \end{pmatrix}$$

Inductive step: We must show that the statement is true for $n = k + 1$. For simplicity, let

$$a = (-1)^k, \qquad b = -k(-1)^k, \qquad c = \frac{k(k-1)}{2}(-1)^k$$

Then

$$\mathbf{A}^k = \begin{pmatrix} a & b & c \\ 0 & a & b \\ 0 & 0 & a \end{pmatrix}$$

Now multiply by $\mathbf{A}$:

$$\mathbf{A}^{k+1} = \mathbf{A}^k\mathbf{A} = \begin{pmatrix} a & b & c \\ 0 & a & b \\ 0 & 0 & a \end{pmatrix}\begin{pmatrix} -1 & 1 & 0 \\ 0 & -1 & 1 \\ 0 & 0 & -1 \end{pmatrix}$$

$$= \begin{pmatrix} -a & a-b & b-c \\ 0 & -a & a-b \\ 0 & 0 & -a \end{pmatrix}$$

Now simplify each entry.

First, the entries on the leading diagonal are

$$-a = -(-1)^k = (-1)^{k+1}$$

Next, the entries just above the leading diagonal are

$$\begin{aligned} a - b &= (-1)^k - \big(-k(-1)^k\big) \\ &= (-1)^k + k(-1)^k \\ &= (k+1)(-1)^k \\ &= -(k+1)(-1)^{k+1} \end{aligned}$$

Finally, the top-right entry is

$$\begin{aligned} b - c &= -k(-1)^k - \frac{k(k-1)}{2}(-1)^k \\ &= \left(-k - \frac{k(k-1)}{2}\right)(-1)^k \\ &= -\frac{2k + k(k-1)}{2}(-1)^k \\ &= -\frac{k^2 + k}{2}(-1)^k \\ &= -\frac{k(k+1)}{2}(-1)^k \\ &= \frac{k(k+1)}{2}(-1)^{k+1} \end{aligned}$$

So

$$\mathbf{A}^{k+1} = \begin{pmatrix} (-1)^{k+1} & -(k+1)(-1)^{k+1} & \frac{k(k+1)}{2}(-1)^{k+1} \\ 0 & (-1)^{k+1} & -(k+1)(-1)^{k+1} \\ 0 & 0 & (-1)^{k+1} \end{pmatrix}$$

Since

$$\frac{k(k+1)}{2} = \frac{(k+1)\big((k+1)-1\big)}{2}$$

this is exactly the required form, so the statement is true for $n = k+1$.

Conclusion: If the statement is true for $n = k$, then it is true for $n = k+1$. Since it is true for $n = 1$, by the principle of mathematical induction the statement is true for all $n \in \mathbb{Z}^+$.

7. The Fibonacci sequence (F_n) is defined by $F_0 = 0$, $F_1 = 1$ and $F_{n+1} = F_n + F_{n-1}$ for $n \geq 1$.
You are given that

$$\mathbf{M} = \begin{pmatrix} 1 & 1 \\ 1 & 0 \end{pmatrix}$$

Prove by induction that

$$\mathbf{M}^n = \begin{pmatrix} F_{n+1} & F_n \\ F_n & F_{n-1} \end{pmatrix}$$

for all integers $n \geq 1$. **[6]**

Solution

We prove by induction that

$$\mathbf{M}^n = \begin{pmatrix} F_{n+1} & F_n \\ F_n & F_{n-1} \end{pmatrix}$$

for all integers $n \geq 1$.

Base case: $n = 1$. The left-hand side is

$$\mathbf{M}^1 = \mathbf{M} = \begin{pmatrix} 1 & 1 \\ 1 & 0 \end{pmatrix}$$

The recurrence gives

$$F_2 = F_1 + F_0 = 1 + 0 = 1$$

So

$$\begin{pmatrix} 1 & 1 \\ 1 & 0 \end{pmatrix} = \begin{pmatrix} F_2 & F_1 \\ F_1 & F_0 \end{pmatrix}$$

Hence

$$\mathbf{M} = \begin{pmatrix} F_2 & F_1 \\ F_1 & F_0 \end{pmatrix}$$

So the statement is true for $n = 1$.

Inductive hypothesis: Assume the statement is true for $n = k$, where k is an integer with $k \geq 1$. That is, assume

$$\mathbf{M}^k = \begin{pmatrix} F_{k+1} & F_k \\ F_k & F_{k-1} \end{pmatrix}$$

Inductive step: We must show that the statement is true for $n = k + 1$.

Using $\mathbf{M}^{k+1} = \mathbf{M}^k\mathbf{M}$ and the inductive hypothesis:

$$\begin{aligned} \mathbf{M}^{k+1} &= \mathbf{M}^k\mathbf{M} \\ &= \begin{pmatrix} F_{k+1} & F_k \\ F_k & F_{k-1} \end{pmatrix}\begin{pmatrix} 1 & 1 \\ 1 & 0 \end{pmatrix} \\ &= \begin{pmatrix} F_{k+1} \times 1 + F_k \times 1 & F_{k+1} \times 1 + F_k \times 0 \\ F_k \times 1 + F_{k-1} \times 1 & F_k \times 1 + F_{k-1} \times 0 \end{pmatrix} \\ &= \begin{pmatrix} F_{k+1} + F_k & F_{k+1} \\ F_k + F_{k-1} & F_k \end{pmatrix} \end{aligned}$$

The Fibonacci recurrence gives

$$F_{k+2} = F_{k+1} + F_k \qquad \text{and} \qquad F_{k+1} = F_k + F_{k-1}$$

So

$$\mathbf{M}^{k+1} = \begin{pmatrix} F_{k+2} & F_{k+1} \\ F_{k+1} & F_k \end{pmatrix}$$

This is exactly the required form, so the statement is true for $n = k + 1$.

Conclusion: If the statement is true for $n = k$, then it is true for $n = k + 1$. Since it is true for $n = 1$, by the principle of mathematical induction the statement is true for all integers $n \geq 1$.

8. The matrix $\mathbf{A}$ is given by

$$\mathbf{A} = \begin{pmatrix} 1 & 1 & 0 \\ 0 & 2 & 1 \\ 0 & 0 & 2 \end{pmatrix}$$

Prove by induction that, for all positive integers n,

$$\mathbf{A}^n = \begin{pmatrix} 1 & 2^n - 1 & 1 + (n-2)2^{n-1} \\ 0 & 2^n & n\,2^{n-1} \\ 0 & 0 & 2^n \end{pmatrix}$$ **[6]**

Solution

We prove the result by induction on n.

Base case: $n = 1$. The claimed formula gives

$$\mathbf{A}^1 = \begin{pmatrix} 1 & 2^1 - 1 & 1 + (1-2)2^{1-1} \\ 0 & 2^1 & 1 \times 2^{1-1} \\ 0 & 0 & 2^1 \end{pmatrix} = \begin{pmatrix} 1 & 1 & 0 \\ 0 & 2 & 1 \\ 0 & 0 & 2 \end{pmatrix} = \mathbf{A}$$

So the statement is true for $n = 1$.

Inductive hypothesis: Assume the statement is true for $n = k$, where k is a positive integer. That is, assume

$$\mathbf{A}^k = \begin{pmatrix} 1 & 2^k - 1 & 1 + (k-2)2^{k-1} \\ 0 & 2^k & k\,2^{k-1} \\ 0 & 0 & 2^k \end{pmatrix}$$

Inductive step: We must show that the statement is true for $n = k + 1$.

Using $\mathbf{A}^{k+1} = \mathbf{A}^k\mathbf{A}$ and the inductive hypothesis:

$$\mathbf{A}^{k+1} = \begin{pmatrix} 1 & 2^k - 1 & 1 + (k-2)2^{k-1} \\ 0 & 2^k & k\,2^{k-1} \\ 0 & 0 & 2^k \end{pmatrix} \begin{pmatrix} 1 & 1 & 0 \\ 0 & 2 & 1 \\ 0 & 0 & 2 \end{pmatrix}$$

Now multiply row by column:

$$\mathbf{A}^{k+1} = \begin{pmatrix} 1 & 1 + 2(2^k - 1) & (2^k - 1) + 2\big(1 + (k-2)2^{k-1}\big) \\ 0 & 2^{k+1} & 2^k + 2k\,2^{k-1} \\ 0 & 0 & 2^{k+1} \end{pmatrix}$$

Now simplify each entry:

$$1 + 2(2^k - 1) = 1 + 2^{k+1} - 2 = 2^{k+1} - 1$$

$$(2^k - 1) + 2\big(1 + (k-2)2^{k-1}\big) = 2^k - 1 + 2 + (k-2)2^k$$

$$= 1 + (k-1)2^k$$

$$2^k + 2k\,2^{k-1} = 2^k + k\,2^k = (k+1)2^k$$

So

$$\mathbf{A}^{k+1} = \begin{pmatrix} 1 & 2^{k+1} - 1 & 1 + (k-1)2^k \\ 0 & 2^{k+1} & (k+1)2^k \\ 0 & 0 & 2^{k+1} \end{pmatrix}$$

This is exactly the required form with $n = k + 1$, since

$$1 + \big((k+1) - 2\big)2^{(k+1)-1} = 1 + (k-1)2^k$$

and

$$(k+1)2^{(k+1)-1} = (k+1)2^k$$

so the statement is true for $n = k + 1$.

Conclusion: If the statement is true for $n = k$, then it is true for $n = k + 1$. Since it is true for $n = 1$, by the principle of mathematical induction the statement is true for all positive integers n.